Raspberry Pi Pico

+3.3V
R1 220
3V1 green
GND

R10 220k
+BATT
+3.3V
GND
J1
5V
NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

P1
rpi pico 1
2648-SC0915CT-ND
tht
3V3_EN
RUN
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
GP8
GP9
GP10
GP11
GP12
bt1
bt2
bt3
bt4
VBUS
VSYS
3V3_OUT
ADC_VREF
GP13
GP14
GP15
GP16
GP17
GP18
GP19
GP20
GP21
GP22
GP26
GP27
GP28
AGND
GND
GND__1
GND__2
GND__3
GND__4
GND__5
GND__6
sda
scl
right_2
right_1
right_en
left_2
left_en
left_1

5V
D1 D_Schottky
+3.3V
R2 220
D13 blue
GND
+3.3V
J3 joystick
GND
+3.3V
J4 speed knob
GND

left motor

U1
DRV8245HQPWPRQ
LVL6, in independent mode that means faults are retries
left_2
left_1
+BATT
GND
C1 0.1uF
SR
DIAG
PH/IN2
EN/IN1
DRVOFF
VM
OUT1
OUT2
GND
EPAD
ITRIP
MODE
NFAULT
IPROPI
NSLEEP
8.2k +-10% LVL2 INDEPENDENT
8.2k R7
left_ipropi
left_en
HSL1 Heatsink
J100 0197054001
J101 0197054001
optionally replace J7 to be weelye compatible
ML+
ML-
J7 1017491

U2
DRV8245HQPWPRQ
right_1
right_2
C2 0.1uF
8.2k +-10% LVL2 INDEPENDENT
8.2k R6
right_ipropi
right_en
HSR1 Heatsink
MR-
MR+
J8 1017491
optionally replace J8 to make the PCB weelye compatible
wellye style J104
wellye RX25 style J105

ADC (for measuring battery voltage and motor current)

+3.3V NT1
NetTie_3
U4
TLA2024IRUGR
+3.3V
+BATT
R13 100k
R14 8.2k
GND
TP0 TestPoint
right_ipropi
left_ipropi
ADDR
AIN0
AIN1
AIN2
AIN3
NC
VDD
SCL
SDA
scl
sda
C4 0.1uF
pullup resistors
TP1 TestPoint left_ipropi
TP2 TestPoint right_ipropi
R11 604
R12 604
+3.3V
R15 3.3k
R16 3.3k

qwiic I2C plug
GND
+3.3V
sda
scl
J2 Conn_01x04
MP

weelye compatible header
J11
+BATT
switch
bat-
D2 D_Schottky
Conn_01x10_Pin

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1 ASJ-200-B-HT
B2 ASJ-200-B-HT
B3 ASJ-200-B-HT
B4 ASJ-200-B-HT
bt1
bt2
bt3
bt4
GND

battery power circuit
J102 0197054001
J103 0197054001
optionally replace J5 to be weelye compatible
J5 TB005-762-02BE
bat-
R3 3.3k
SW1 screw terminal
C5 33uf
+BATT
GND
18V zener D3
13.7k R4
switch
J6
PSMN1R2-25YLDX NMOS Q1
reverse polarity protection
PSMN1R2-25YLDX Q2
on/off switch

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1 M3
H2 M3
H3 M3
H4 M3

alternative voltage regulator
Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.
U5
D78B05T, RBT05W24S05, pololu5593
+BATT
IN
GND
OUT
5V

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7F7D983B50C08E2
U3
TPS54202HDDCR
+BATT
R5 100k
C6 0.1uF
L1 12uH
5V
EN
FB
VIN
BOOT
SW
GND
C3 10uF
C7 56pF
R8 100k
R9 13.7k
C8 47uF

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
Sheet: /
File: gbg-pcb.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.2 | Id: 1/1